



IDOR Response Platform

End-to-end Digital Forensics & Incident Response for IDOR investigation

TECHNICAL CHALLENGE — Cybersecurity

GenAI Engineering

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Backlog

Objetivos e Requisitos do projeto



1

Challenge Context

High-volume suspicious activity
Need for automated triage and forensic analysis
Safe dry-run response with human approval

2

Challeng Requirements

Investigation
Detection
Forensics
GenAI
NIST IR
Human Approval
Observability

End-to-end Digital
Forensics & Incident
Response for IDOR
investigation



Data Understanding & Feature Discovery

- 4.478.619 logs
- 5.726 IPs
- 182 IDOR findings
- 172 anomalous IPs
- 586 IOCs

Dataset Challenges

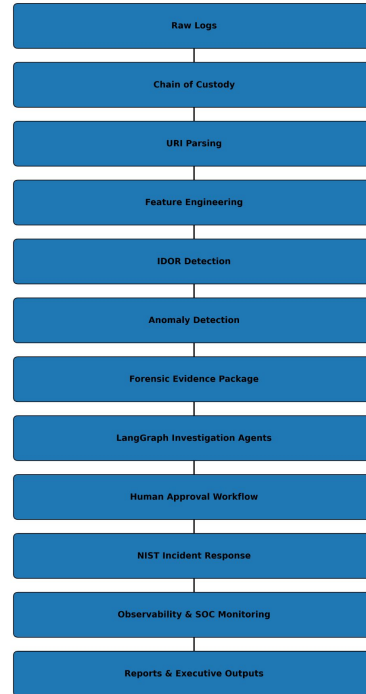
- Scale
- Performance
- Evidence integrity
- Reproducibility



Architecture Overview



IDOR Response Platform Architecture



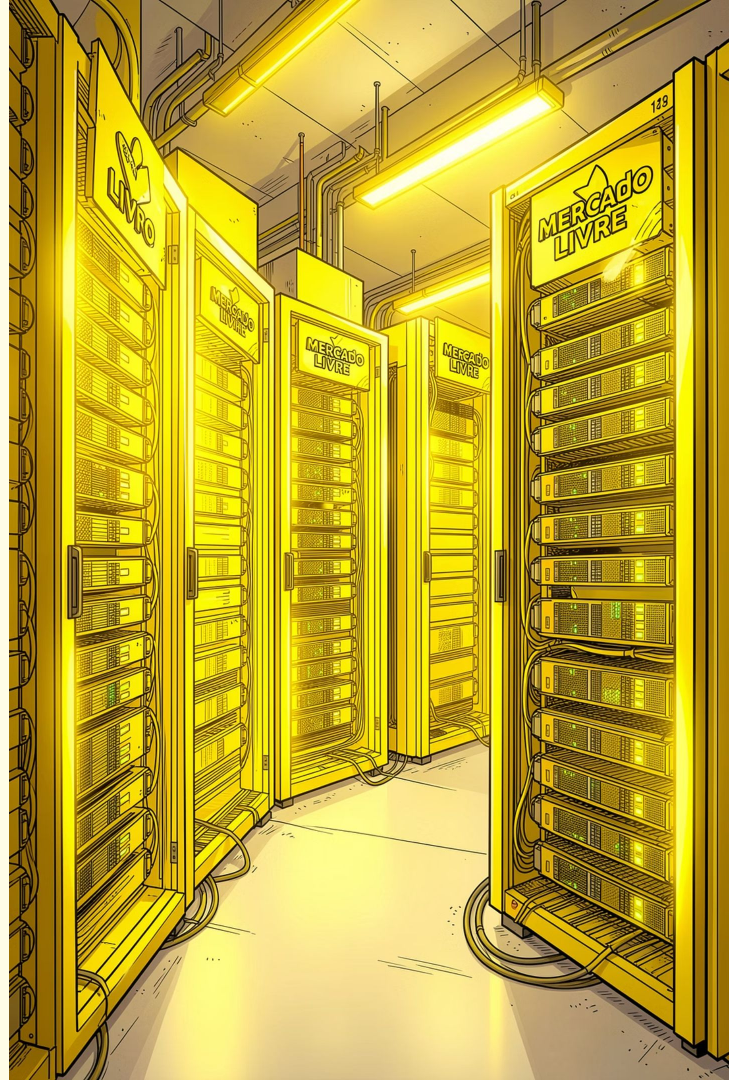
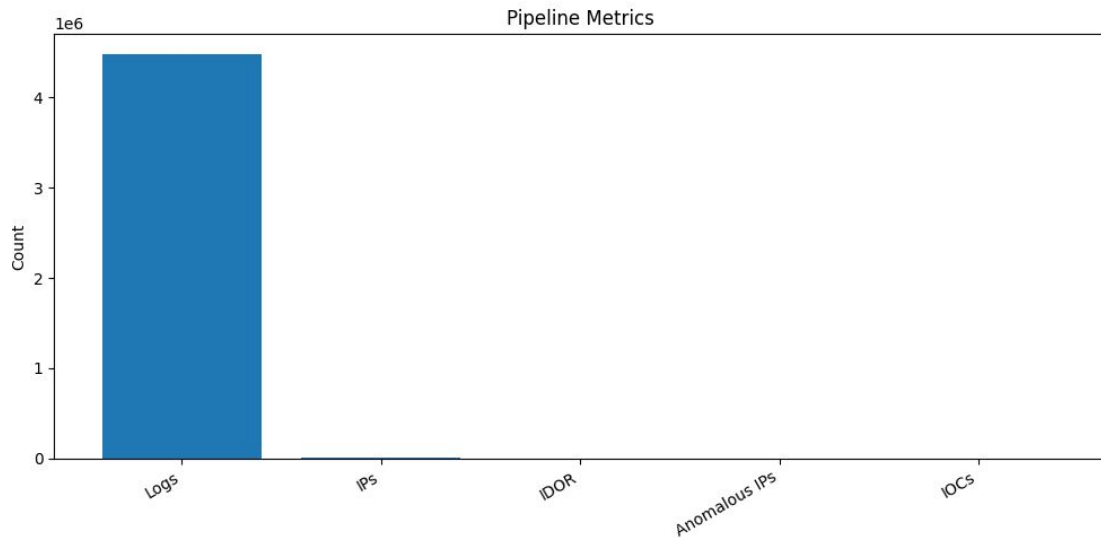
STACK

- Polars
- Parquet
- Pydantic
- Pytest
- LangGraph
- Streamlit

Seis algoritmos avaliados em pipeline padronizado com validação cruzada estratificada. Métricas reportadas: ROC-AUC, PR-AUC, F1 e matriz de confusão — garantindo comparação justa e reprodutível.



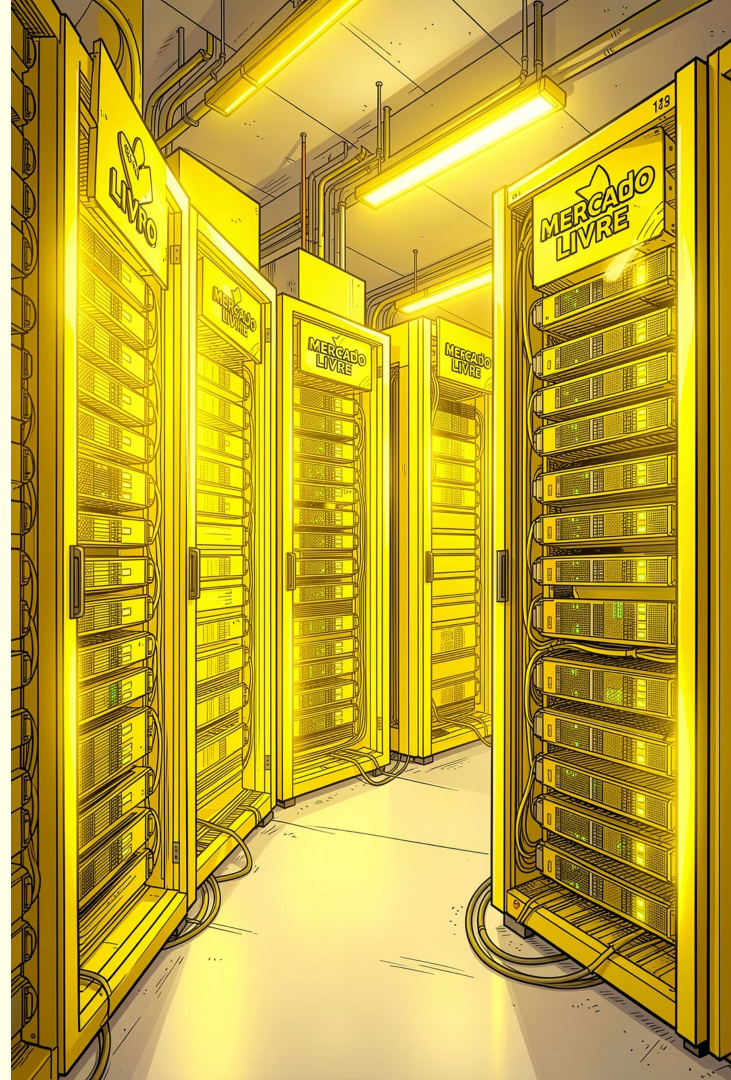
Pipeline Metrics





Data Pipeline

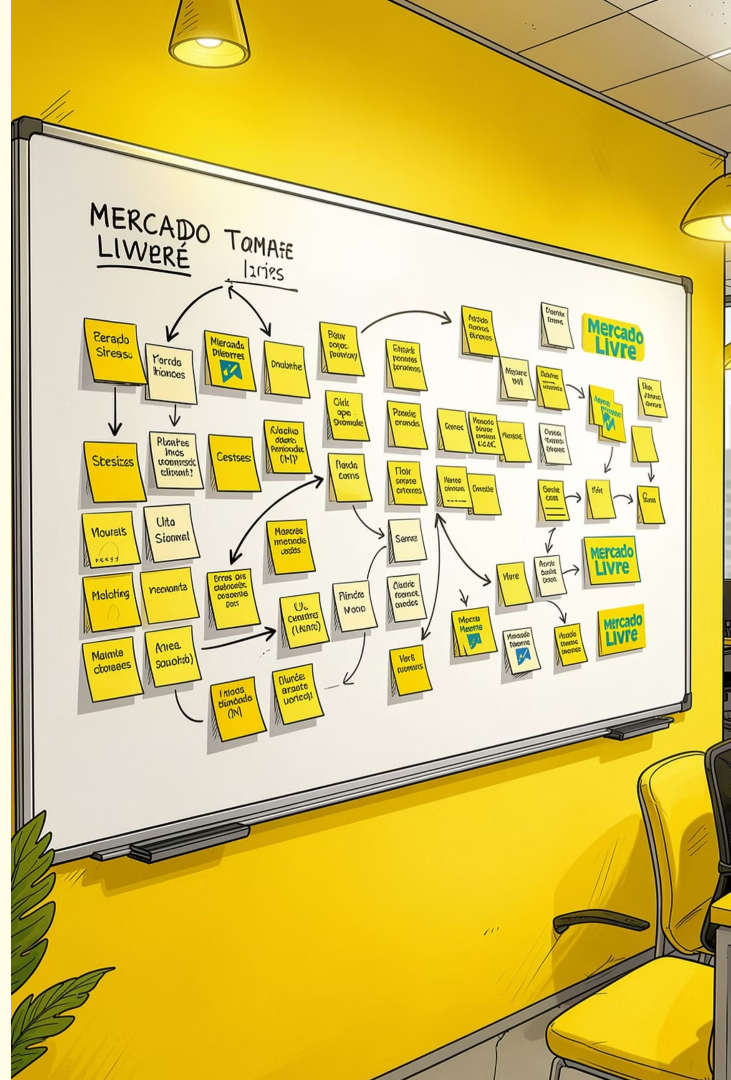
- Chain of custody
- URI parsing
- Feature engineering
- Parquet artifacts





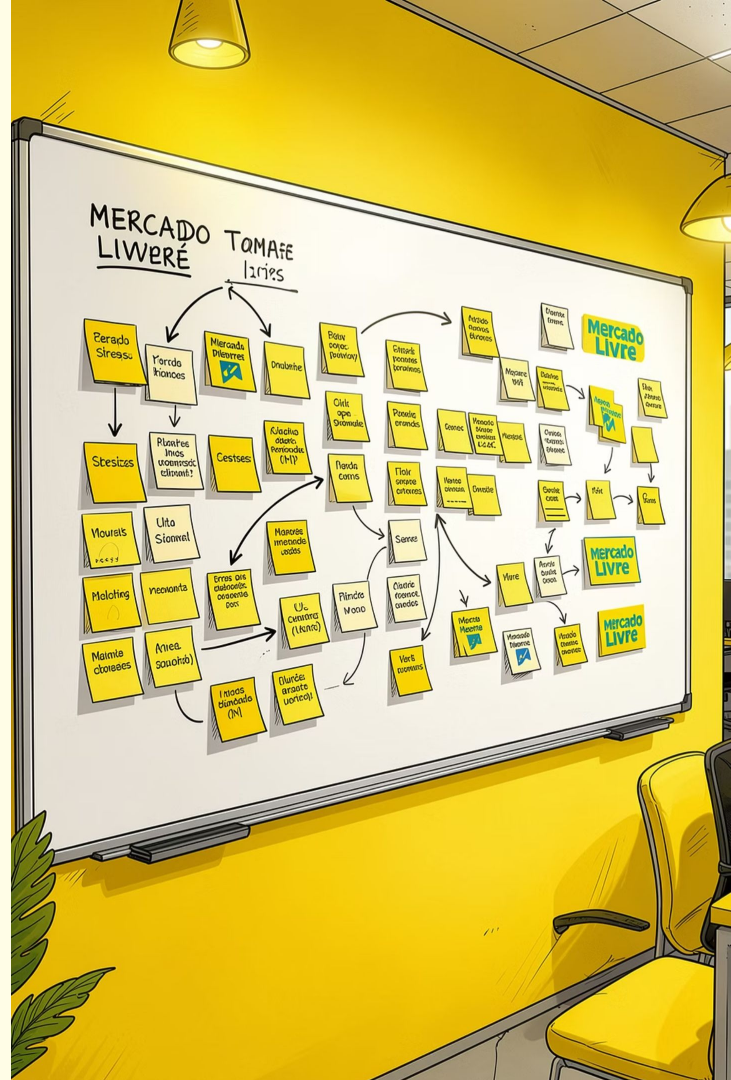
IDOR Detection

- Invoice diversity
- Sequential access
- Token reuse
- Endpoint abuse





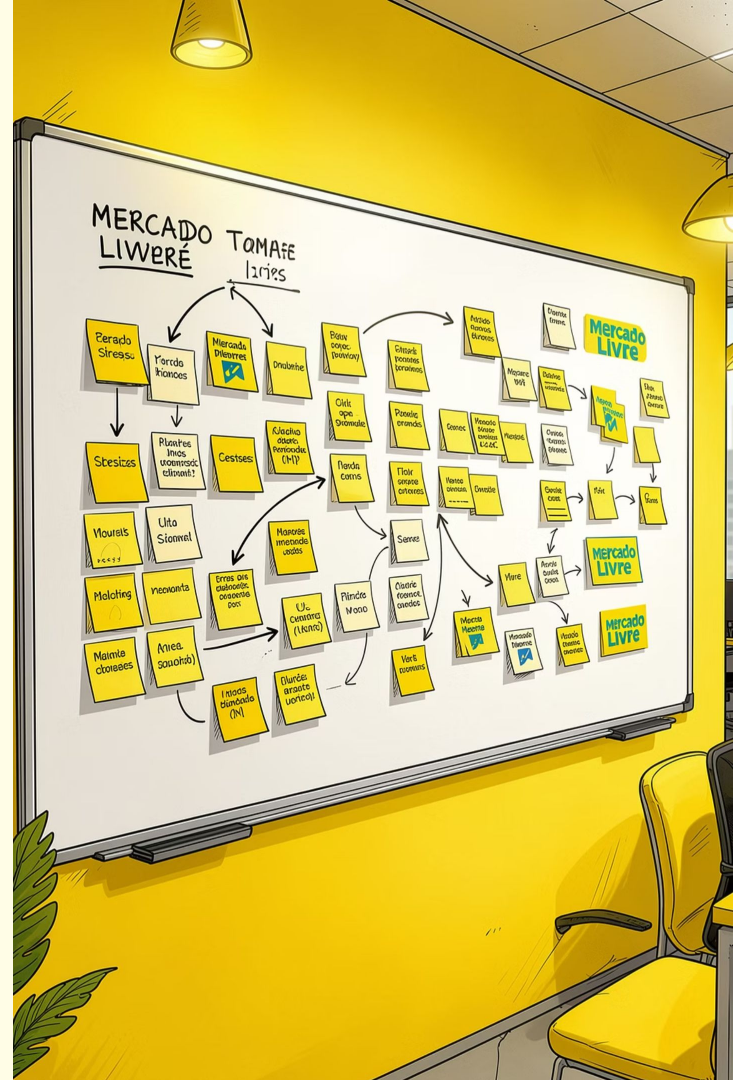
- Prioritization
- Analyst focus
- Evidence-driven



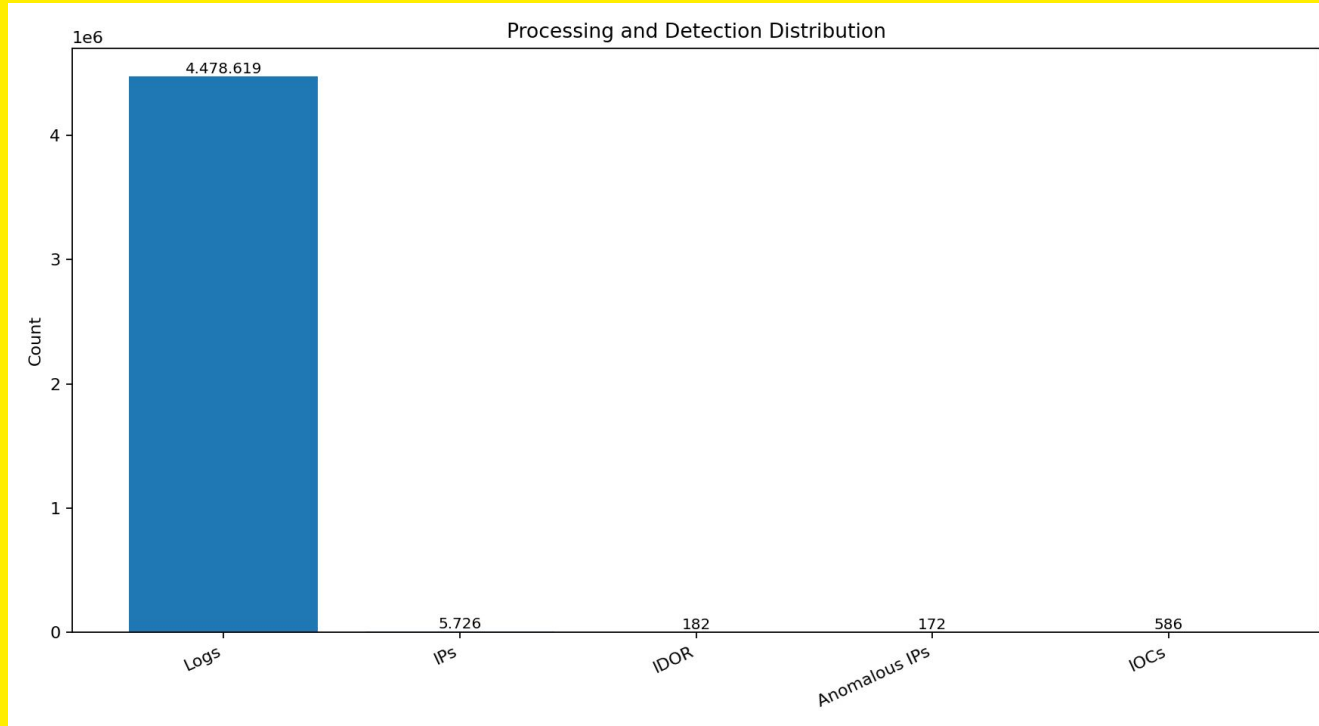


Anomaly Detection

- Isolation Forest
- 172 anomalous IPs



Anomaly Detection

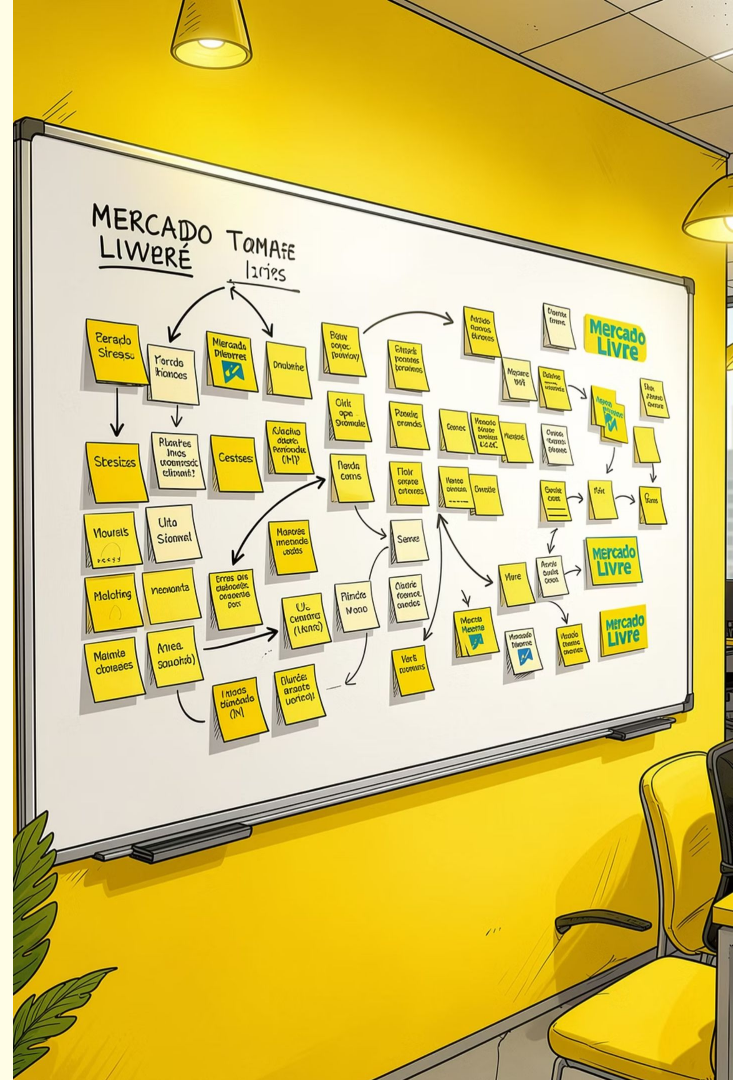


Isolation Forest complements deterministic IDOR logic.



Detection Results

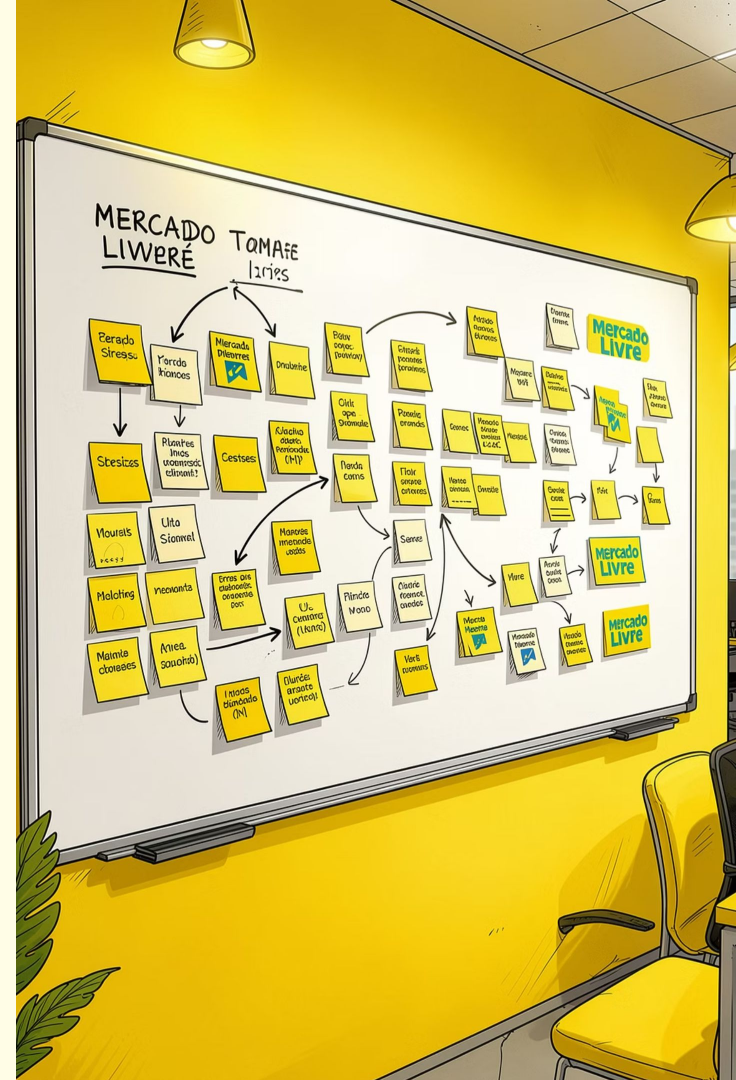
- Detection Results



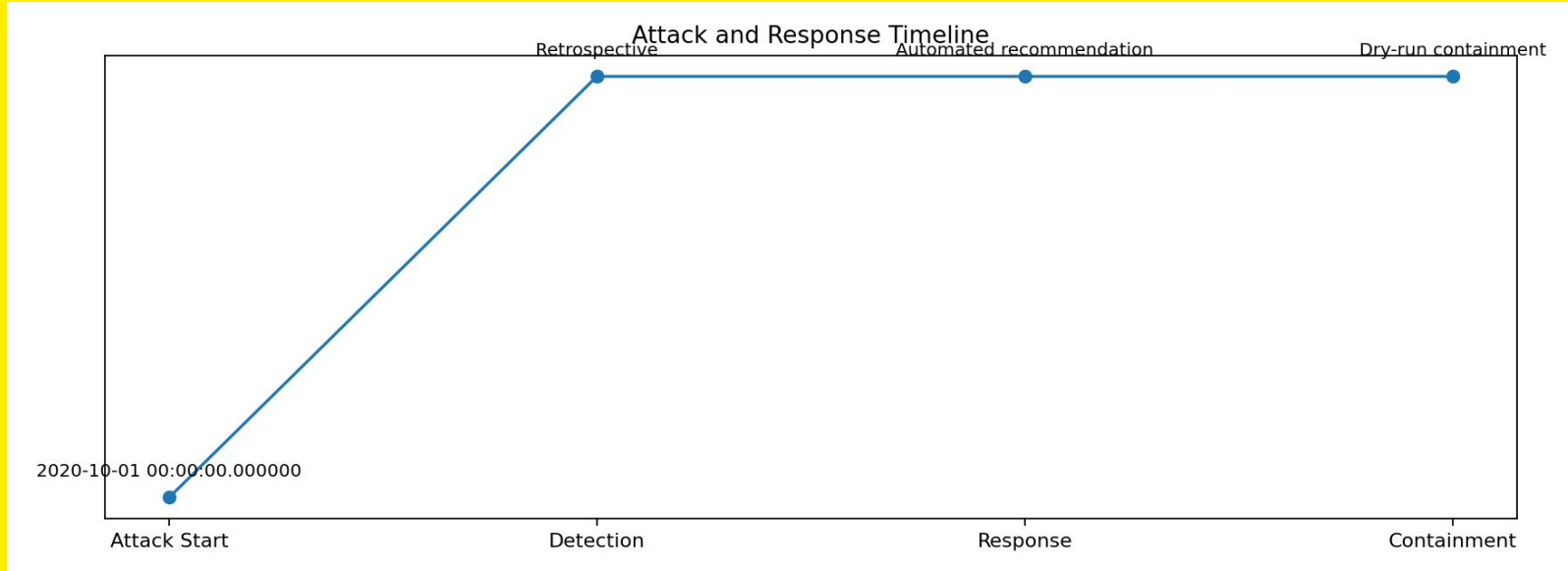


Forensic Investigation

- Patient Zero
- Attack Window
- Impact



Forensic Evidence

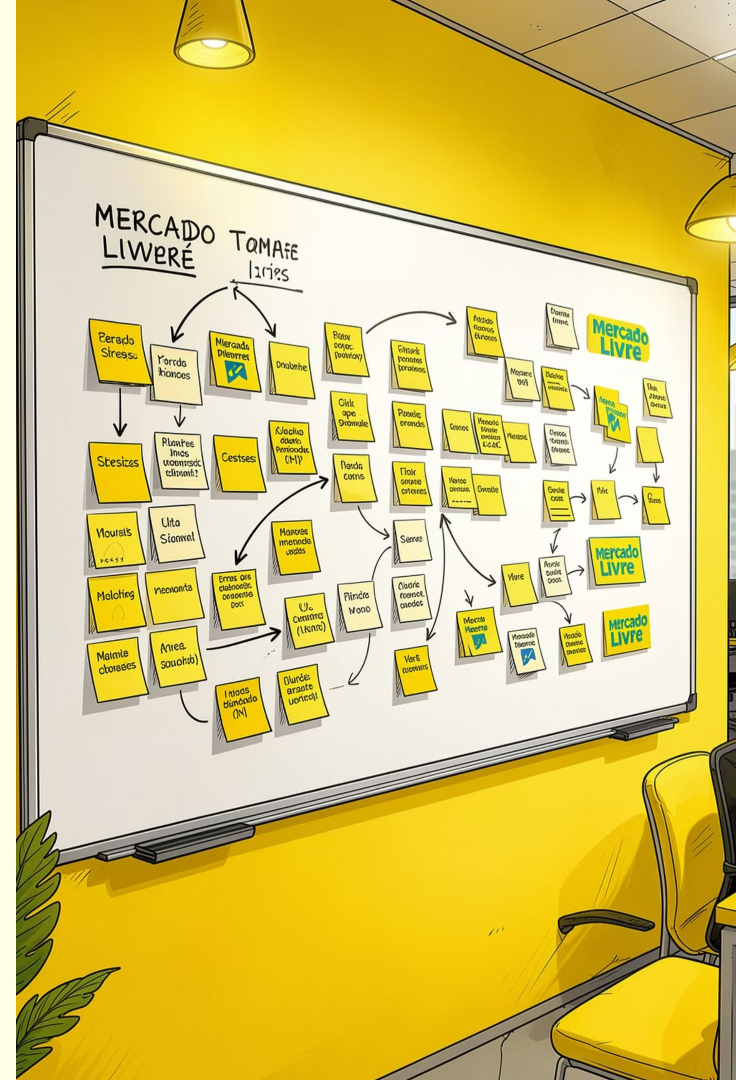


Patient zero, attack window, IOCs and impact.



IOC Generation

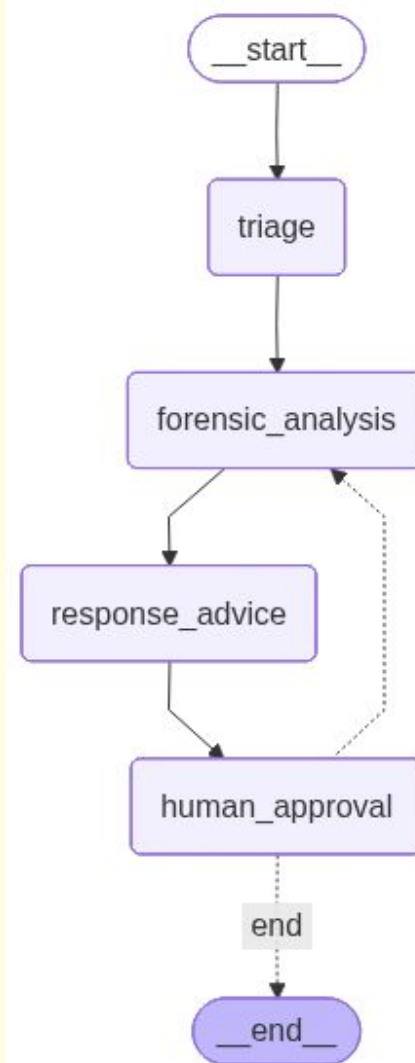
- IPs
- User Agents
- Enumeration Patterns





Agentic Orchestration - LangGraph - MCP

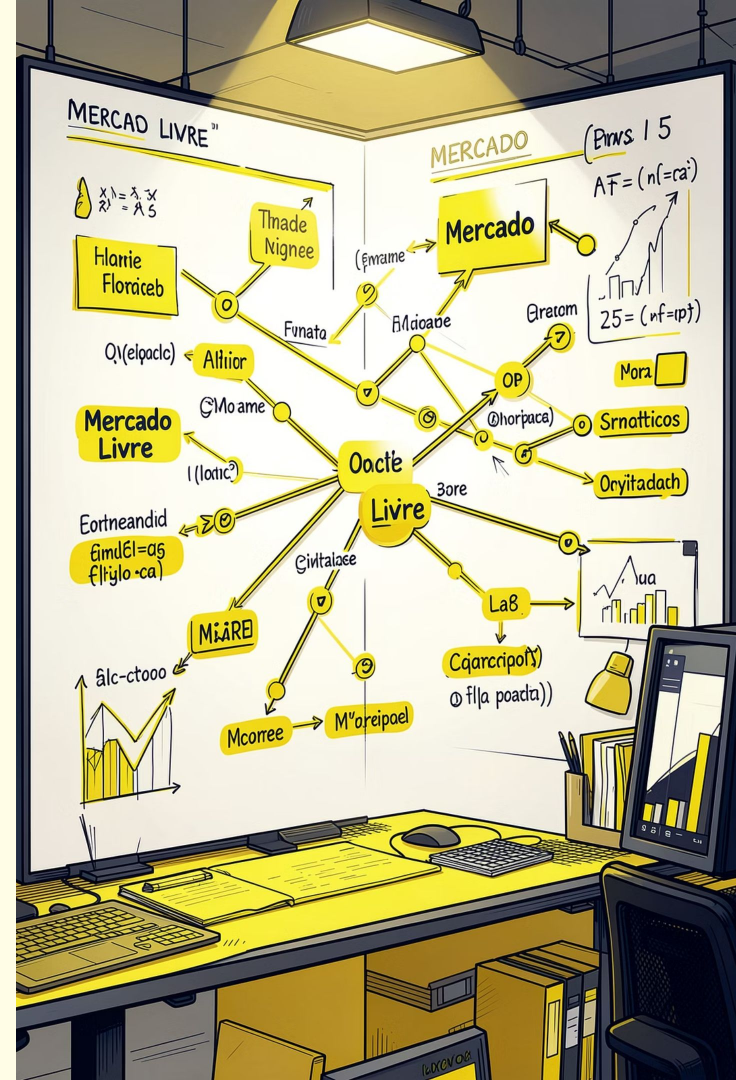
①





Triage Agent

- Severity
- Prioritization
- Hypotheses

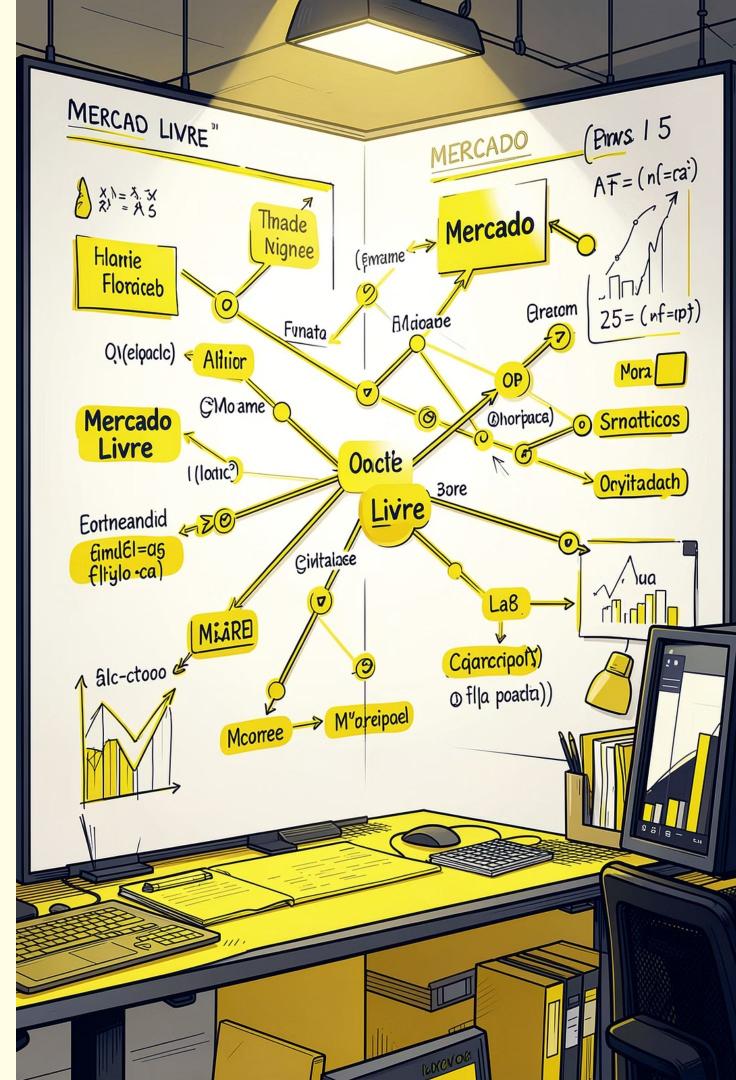




Response Agent

①

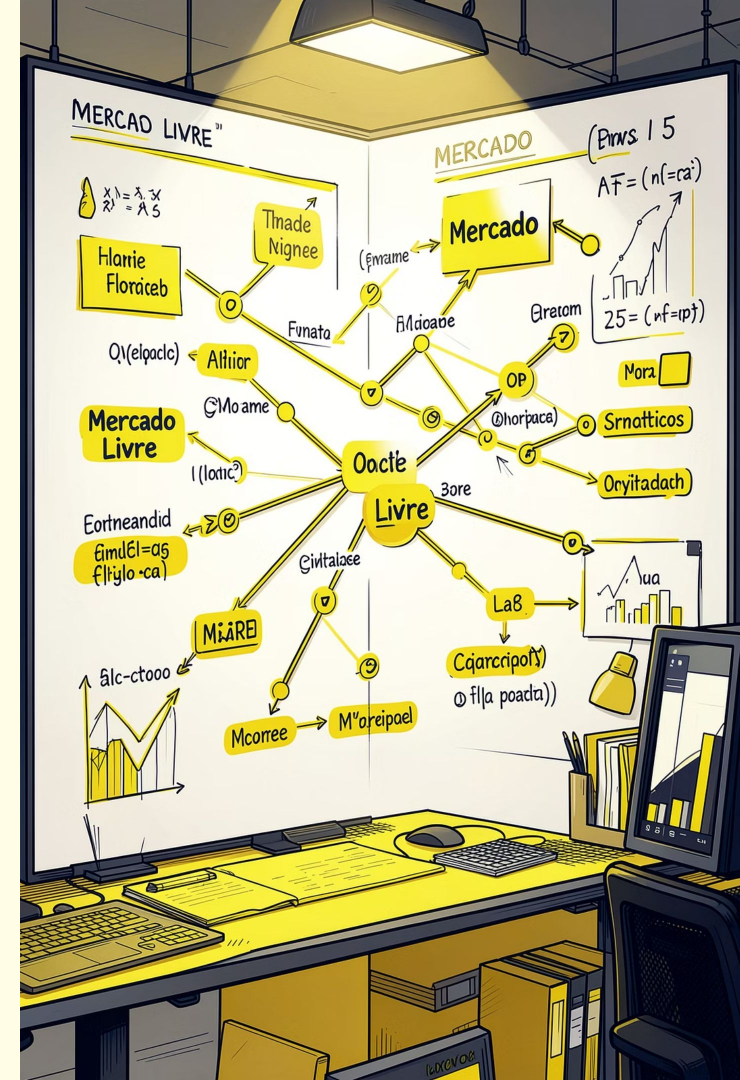
- Containment
- WAF
- Rate limiting





Human Approval

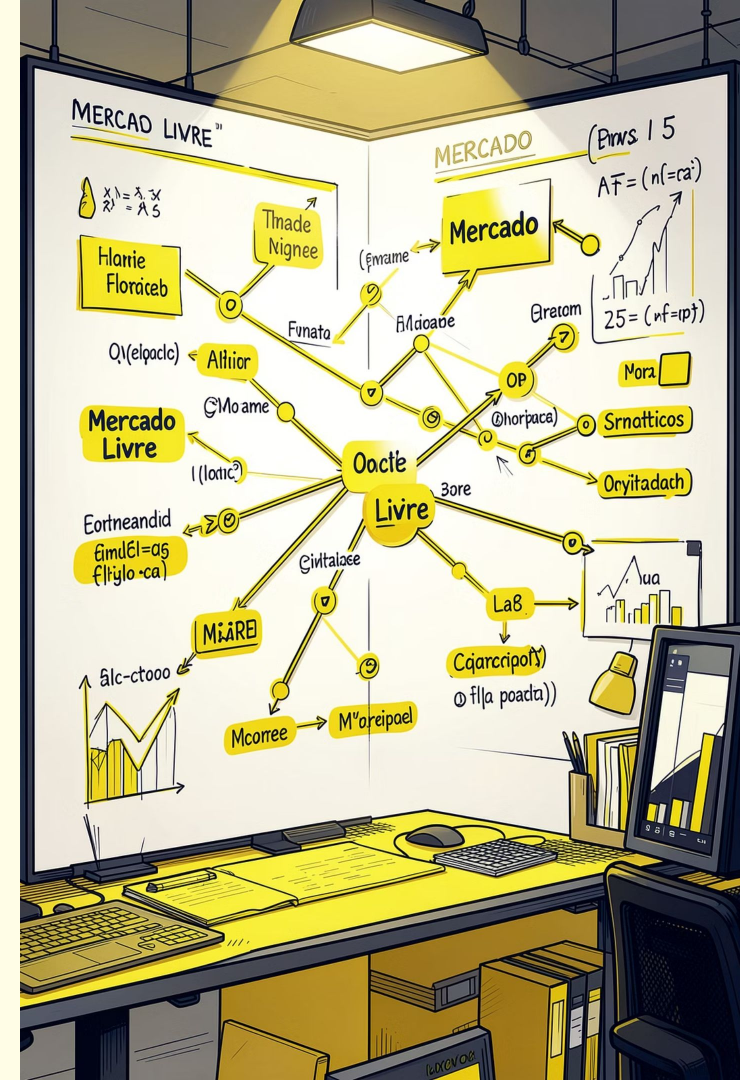
- Approve
- Reject
- Modify
- Request More Evidence



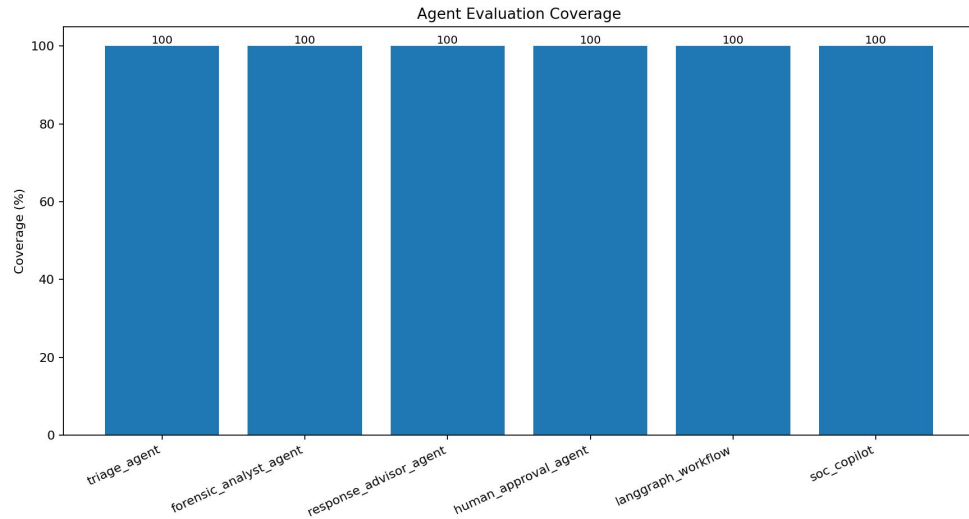


Human Approval

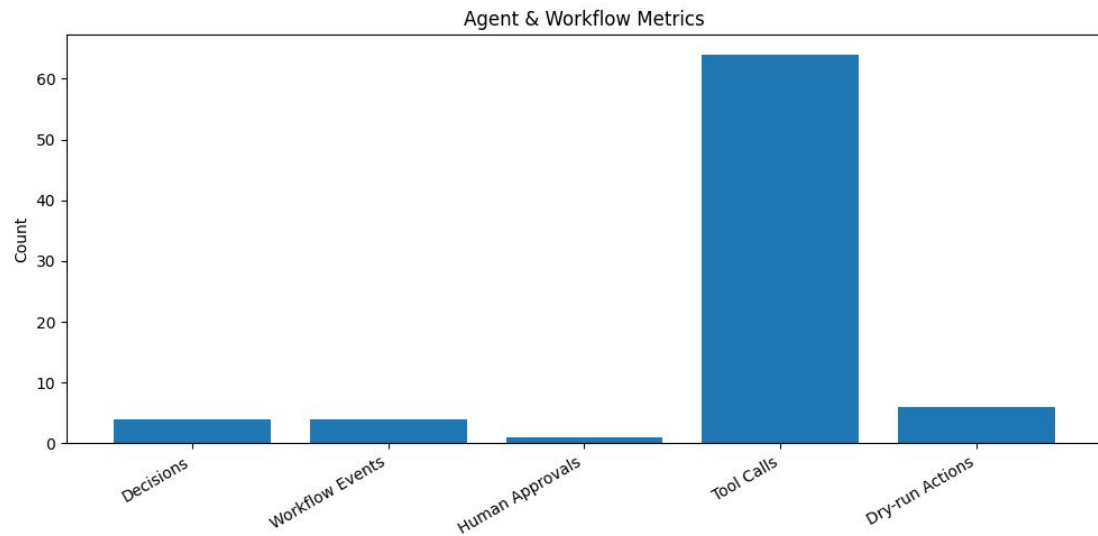
- Approve
- Reject
- Modify
- Request More Evidence



Agent Evaluation



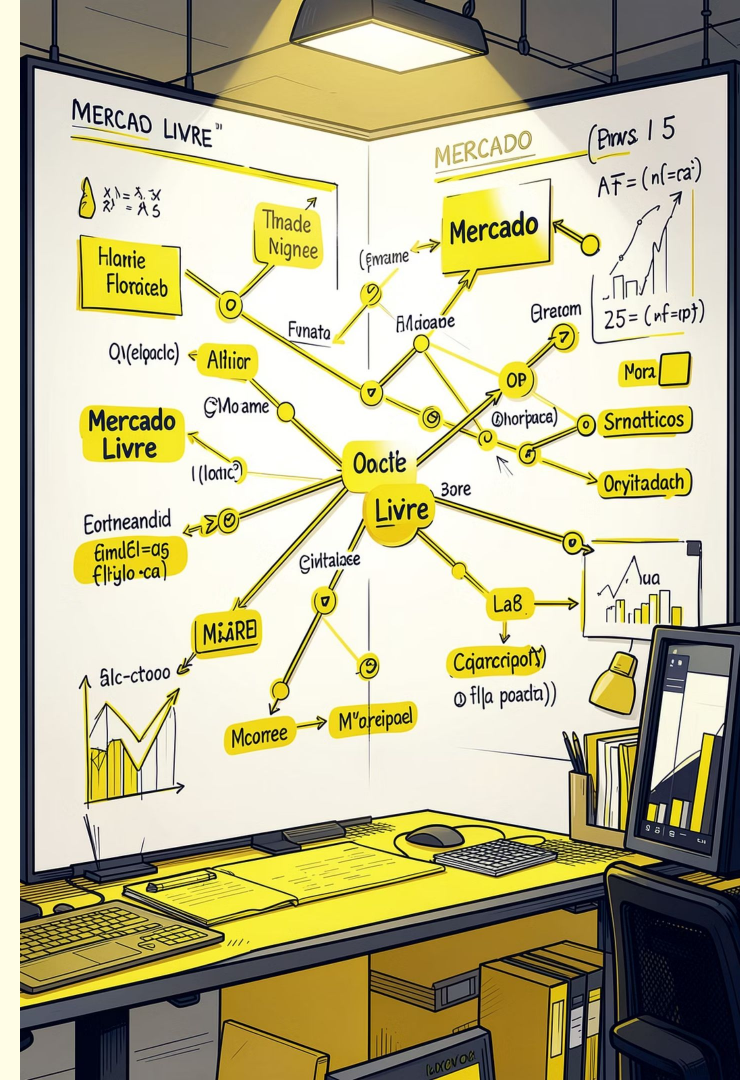
100% agent evaluation coverage.



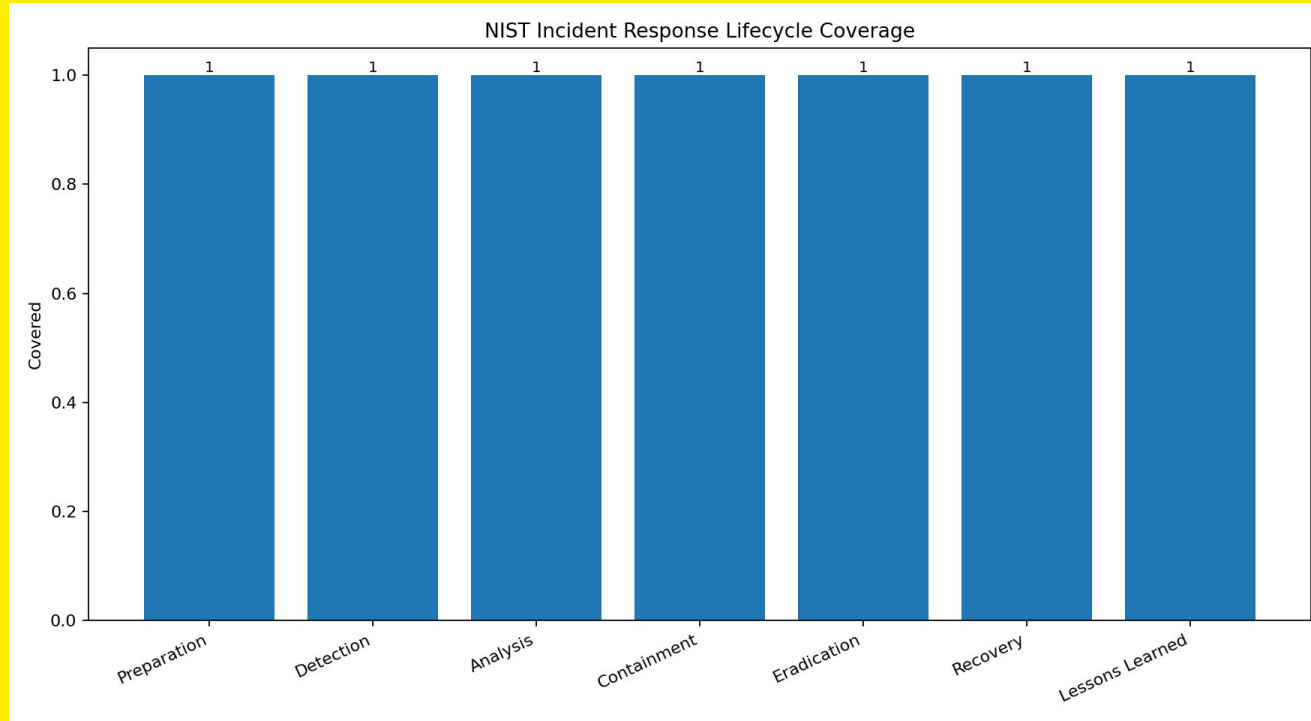


NIST Incident Response

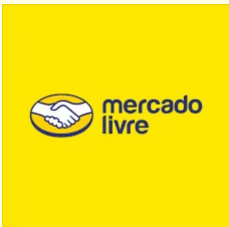
- Detection
- Analysis
- Containment
- Eradication



NIST Response

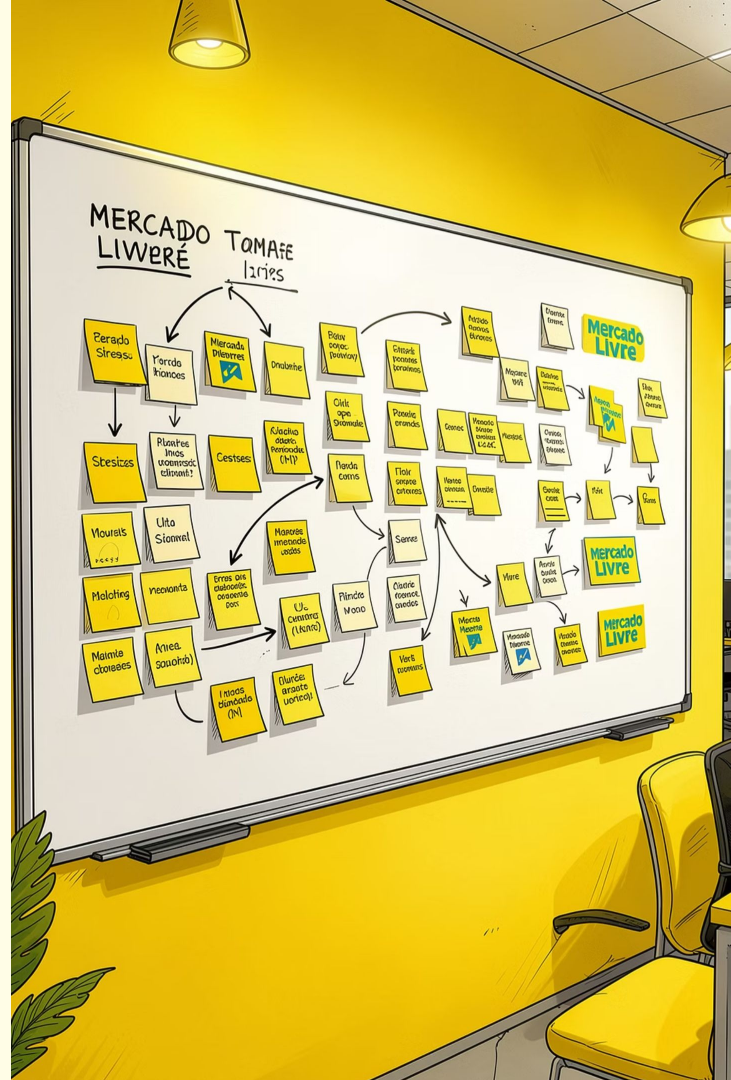


TTD, TTR, TTC, containment, eradication and root cause.

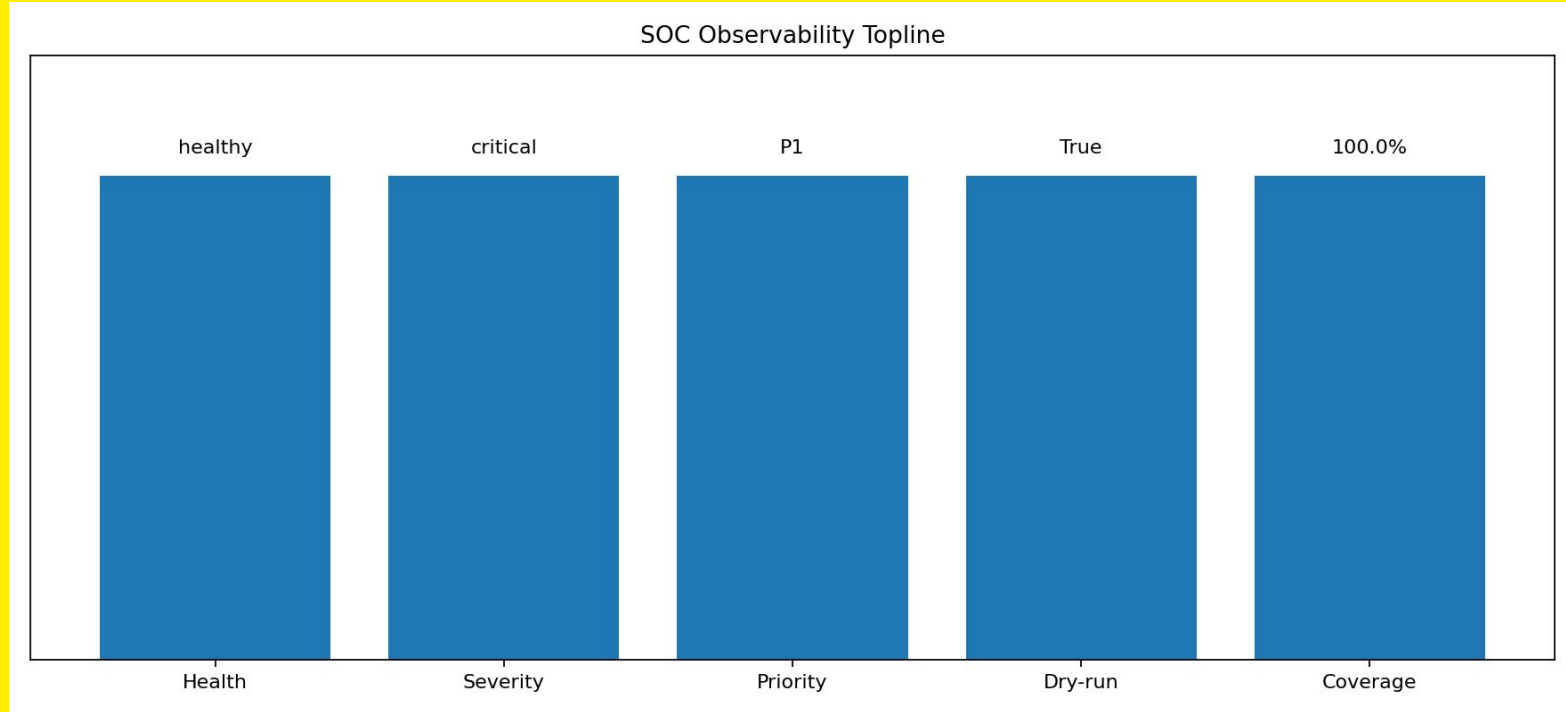


Response Metrics

- TTD
- TTR
- TTC



Observability

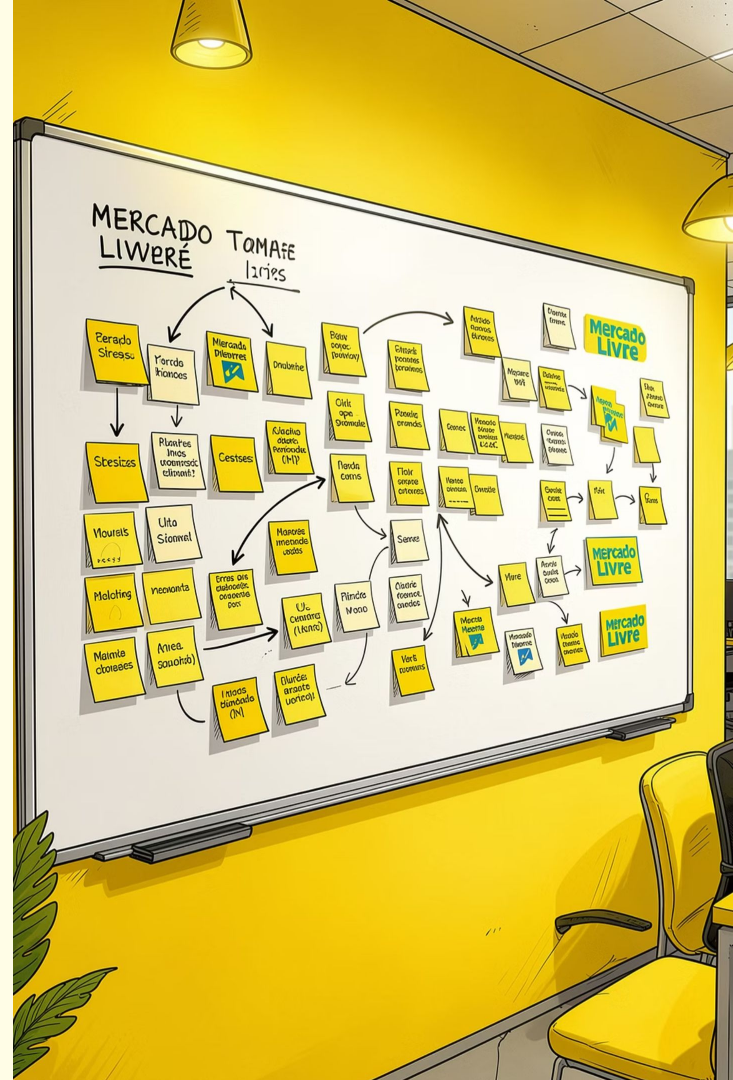


SOC health, severity, priority, coverage and dry-run state.

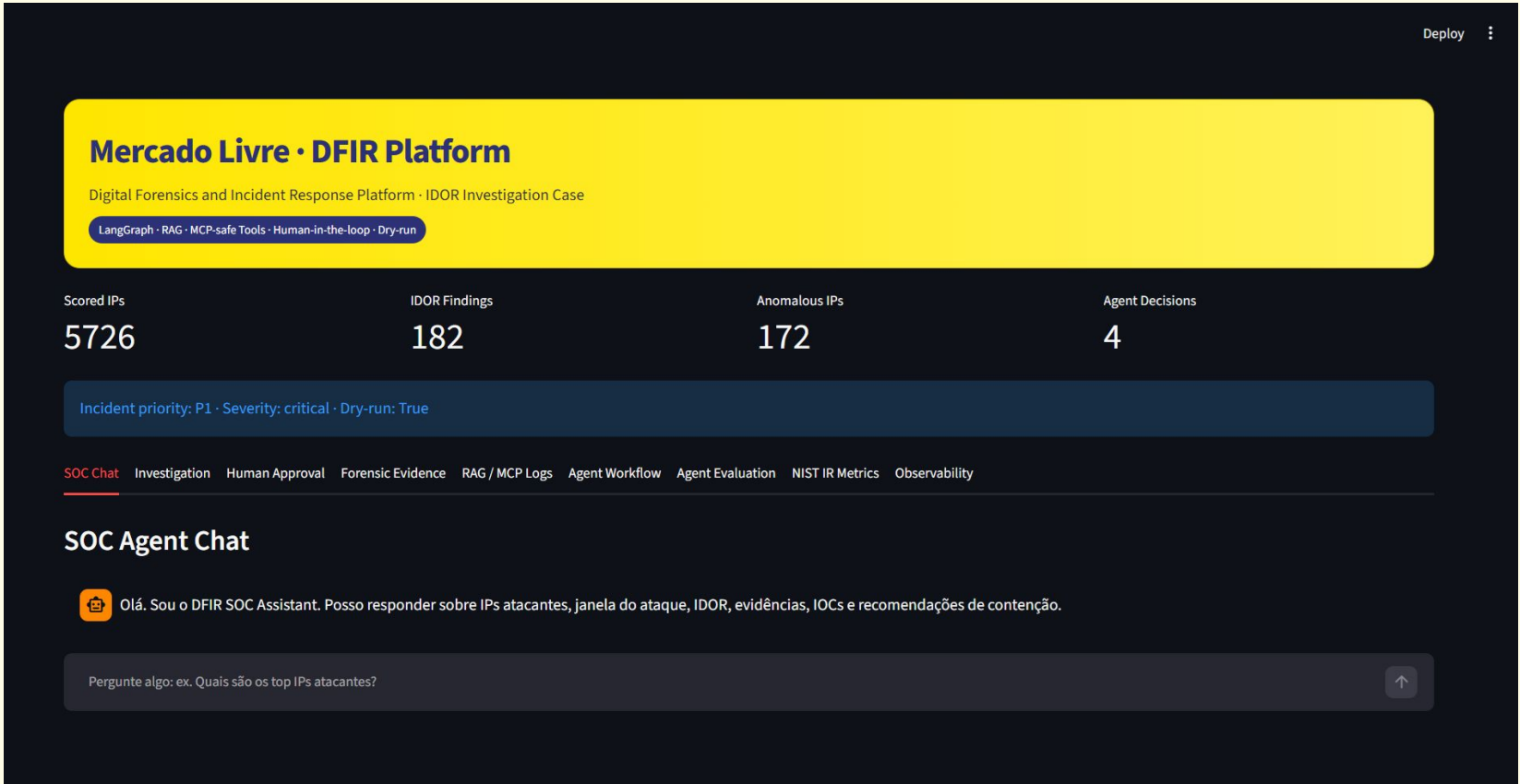


Business Impact

- Reduces manual investigation effort
- Improves evidence quality
- Supports analyst decisions
- Avoids unsafe automatic containment



- Prototype front-end of the platform

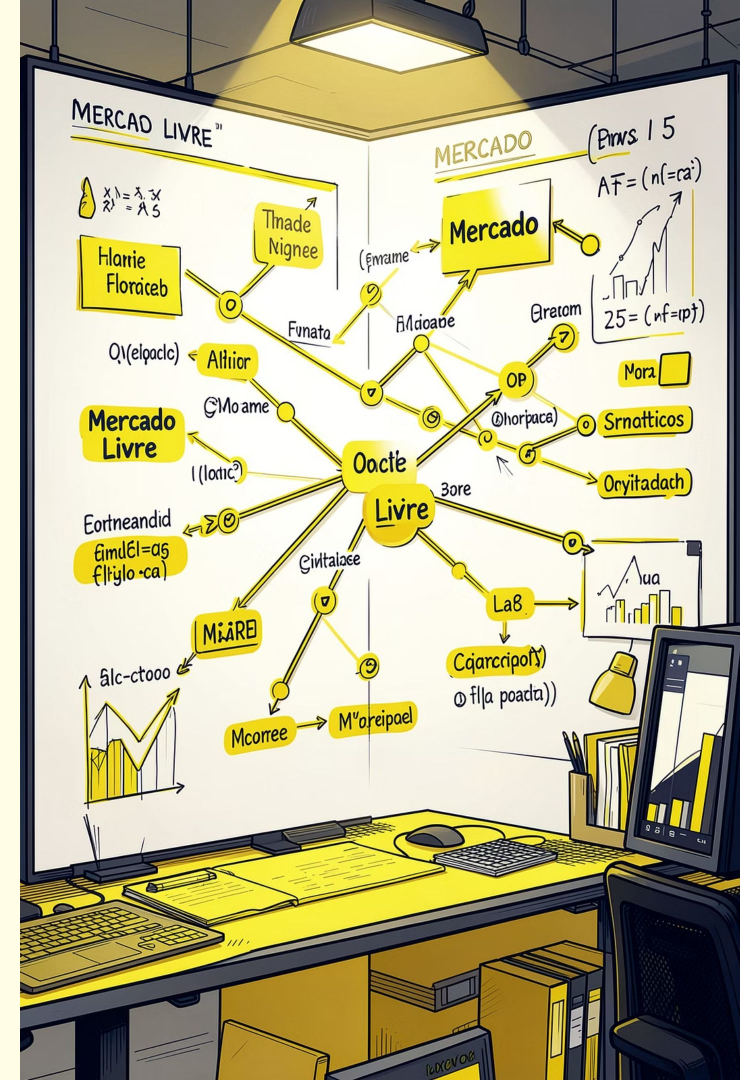




Platform Evolutions (Next Releases)

1

- Docker
- Grafana
- Prometheus
- RBAC
- SIEM





Q&A

Thank you!